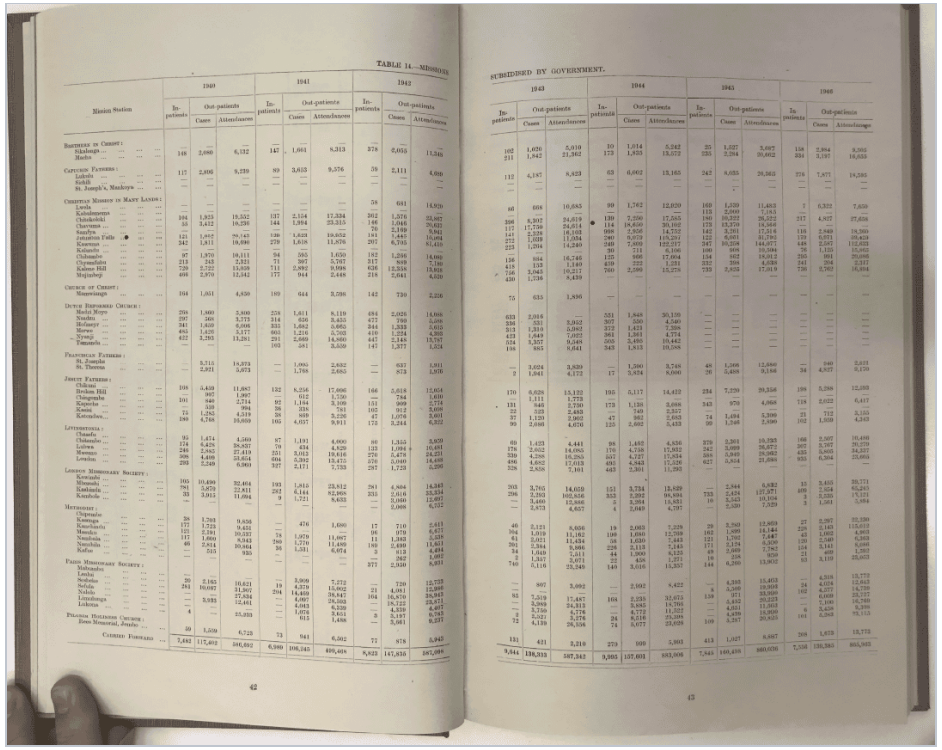
The Long Shadow of Mission Medicine

Colonial missionary healthcare, institutional trust, and the spatial burden of disease in Sub-Saharan Africa

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Wellcome Early-Career Award | 355277/Z/26/Z | January 2027



Northern Rhodesia medical report, Table 14: mission stations subsidised by government, 1940 to 1946

Clinics are within reach, but many people don't use them

The usual explanations

Poverty

Distance

Facility quality

↓

…leave a gap: similar communities still differ in use

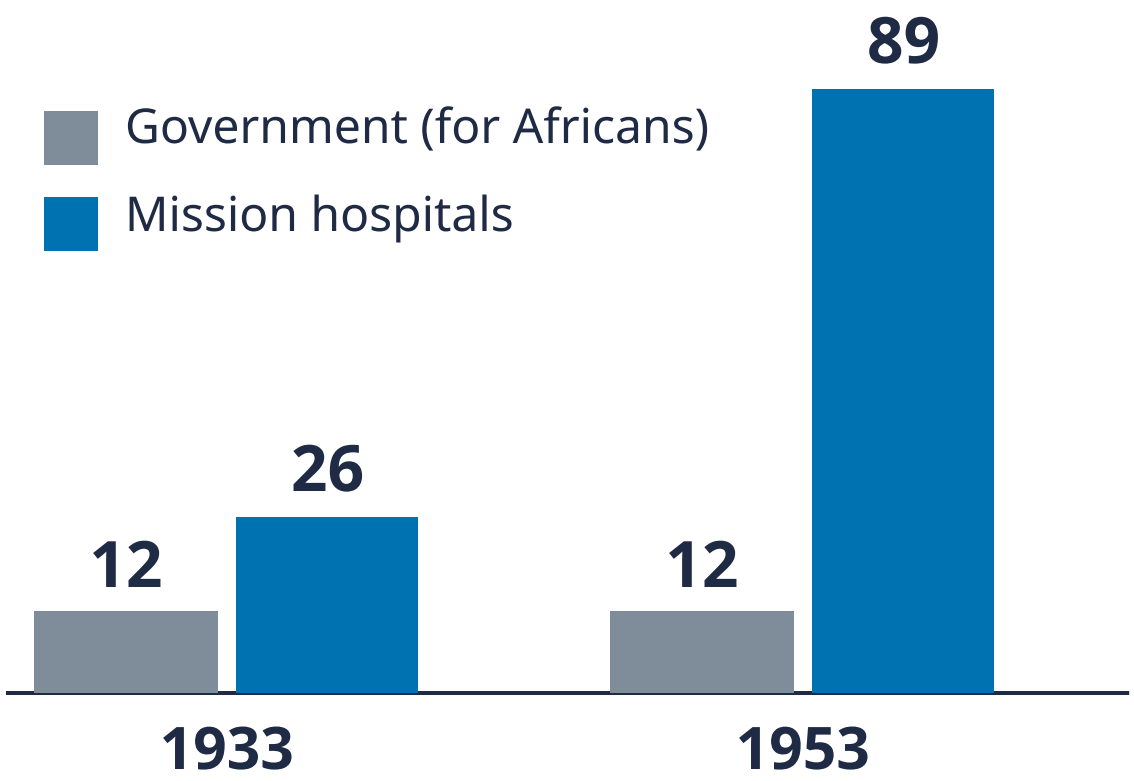
Missing piece: why trust varies

[Insert map: within-country variation in facility births or DTP3 by DHS cluster, Zambia, with health facilities overlaid]

Sources: Peters et al. 2008; Ensor and Cooper 2004; Gilson 2003; Østergaard 2015; Weiss et al. 2019

For decades, missionaries ran most of the Western healthcare Africans met

Hospitals in Northern Rhodesia



Source for chart: Chiseni (2025), African Studies 84(1-2), Figure 3 and text

Trust deficit

Coercive colonial medical campaigns in central Africa still lower trust and vaccination (Lowes and Montero 2021)

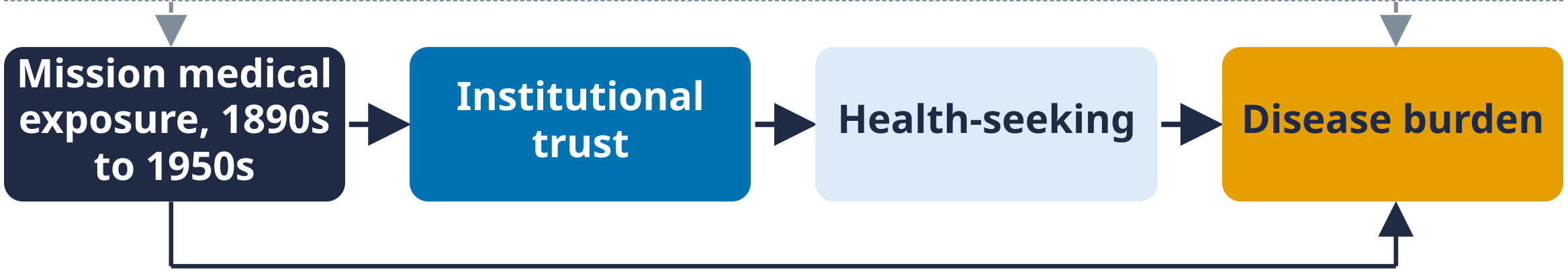
Trust dividend

Living near a colonial medical mission in India still predicts better health (Calvi and Mantovanelli 2018)

My hypothesis: mission medicine left a dividend in some places and a deficit in others

Four questions, one causal chain

I control for where missions chose to go and where the state built hospitals



Direct route: facilities that survive today

RQ1 Map
Where was mission care?

RQ2 Link
Trust, behaviour and burden today

RQ3 Channels
Facilities, health workers, norms, coercion

RQ4 Meaning
How communities see it

I've already built this for Zambia, from records others haven't used

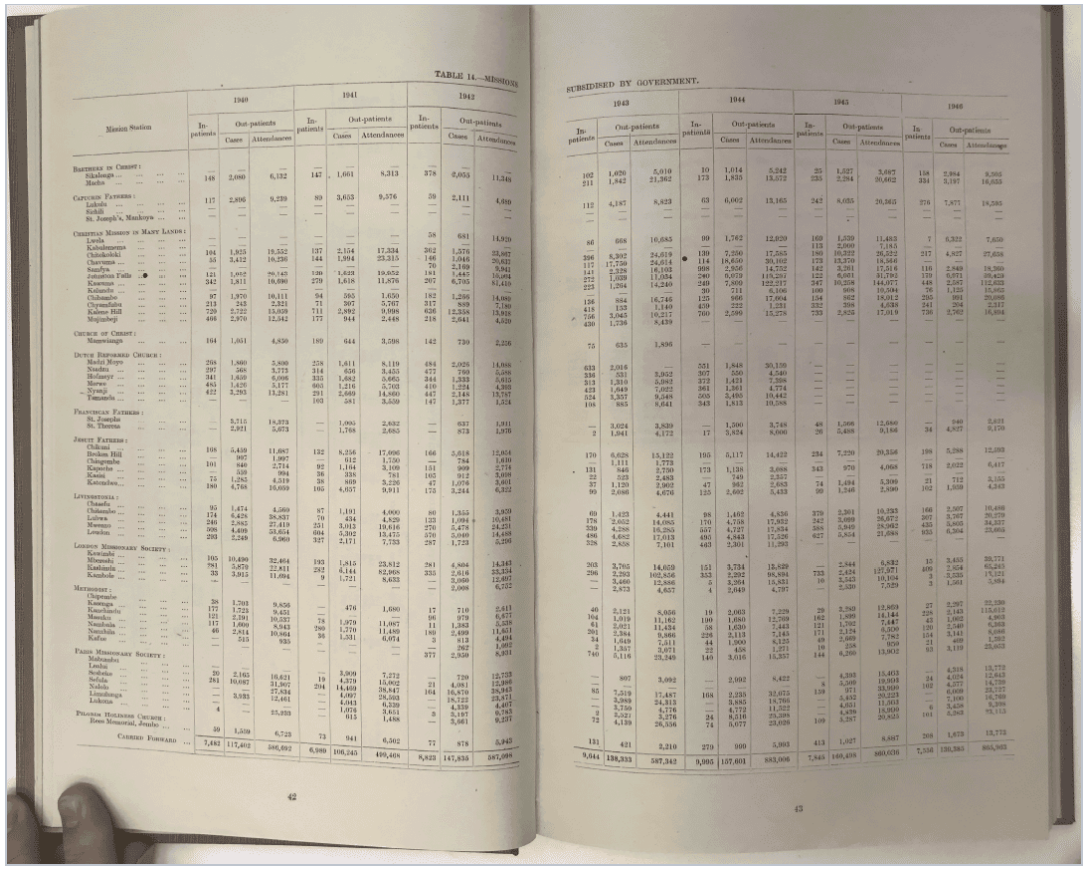


TABLE 14.—MISSIONS SUBSIDISED BY GOVERNMENT.

Northern Rhodesia medical report, Table 14: subsidised mission stations, in- and out-patients, 1940 to 1946

76 to 91%

of missions are missing from the atlases most studies use (Jedwab et al. 2022)

Blue Book church returns

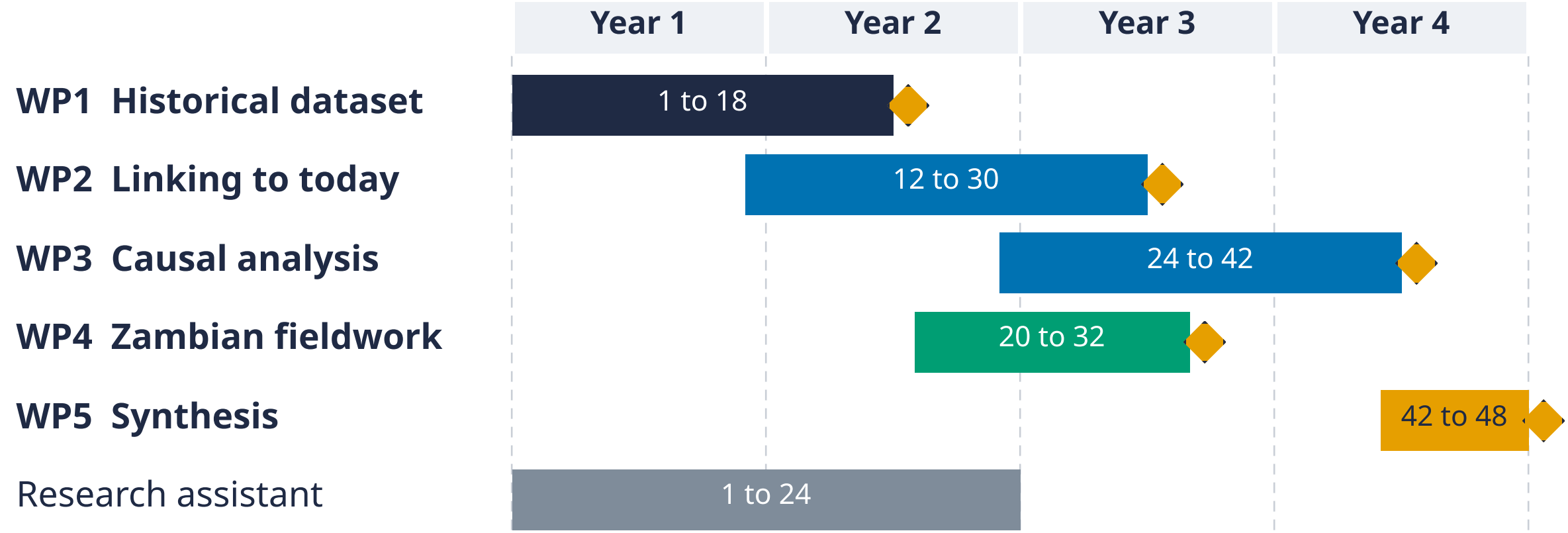
Every mission station, by society

Colonial medical reports

Facility type and yearly patient numbers

= open, facility-level dataset for British Africa

48 months, five work packages



Milestones (diamonds): v1 dataset and data paper (m18), first article (m30), fieldwork coded (m32), burden and mediation article (m42), open dataset and policy briefs (m48)

Separating the effect of mission medicine from where missions chose to go

The objection

Missionaries chose healthier, more accessible, richer places.

A simple comparison would mix up the effect of medicine with the effect of place.

Jedwab, Meier zu Selhausen and Moradi (2022)

1. **Control for location drivers**
   Malaria, railways, cash crops, state hospitals
2. **Medical vs non-medical stations**
   Same choice of location, different medical exposure
3. **Societies' medical propensity**
   Some societies did far more medical work than others
4. **Dose-response and distance decay**
   Effects grow with exposure, fade with distance

Report sensitivity bounds, not a claim of perfect exogeneity

Asking communities what this history means

2
communities with long mission-hospital histories

2
comparison communities without

Matched on today's access to care, province and language

Who we talk to
Elders, health workers, healers, church and traditional leaders, young people

~50 semi-structured interviews (40 to 60)

8 focus groups, women and men separately

30 days in the field with 4 Zambian assistants

1 community feedback visit before publication

What the reviewers asked, and what has changed

Reviewers asked	My response
How is trust measured?	DHS blood-test refusal; trust in the Ministry of Health against police and courts (Afrobarometer R10); fieldwork
Is the mediation credible?	Total effect is primary; trust share via sequential g-estimation with sensitivity bounds
Is continental coverage feasible?	Coverage audit: [N] territories, [years]; core set by month 18

Since submission

Afrobarometer Round 10 (2024, 38 countries) adds a direct question on trust in the Ministry of Health. [Add: advisory names, Zambian partner, coverage audit]

I've already done the hardest part once

2022	PhD, Lund: built the Zambian dataset
2024	Missions and HIV today (EHDR)
2024	Open-access Springer monograph
2024	Gender and schooling (with J. Bolt)
2025	Mission healthcare (African Studies)
Forthcoming	IV design: missions and wealth (Elgar)
Postdoc	Six months of fieldwork in rural Zambia

Sole author: thesis, monograph, EHDR, African Studies, Elgar chapter

This award adds

- Scaling across many archives
- Causal mediation
- Spatial epidemiology
- Leading a team

Stellenbosch gives me the historical-data core; partners bring the health expertise



An African host: close to field sites and policy audiences, and the skills stay on the continent

How I'll run the team

Train the research assistant

Written development plan; archival, GIS and data skills; co-authorship on the data paper when earned

Fair pay and credit in Zambia

Fieldwork staff trained, paid at fair local rates, credited by name

Communities see results first

Feedback visit to review and interpret findings before publication

Open by default

Dataset, code and papers openly available; pre-analysis plan

[Optional: flexible working, from my own experience of parental leave]

If trust is the lever, buildings alone won't close the gap

If I'm right: part of Africa's health inequality is historical, and trust is a lever that infrastructure alone won't move.

Whatever the result: the first open, facility-level map of mission medicine in British Africa.

Thank you

Michael Chanda Chiseni | Stellenbosch University (LEAP)

Backup slides

For questions only. Check with your Funding Manager that backup slides are allowed during Q&A.

Backup: how I measure trust

Revealed

Refusal of free DHS blood tests (Lowes and Montero 2021)

Same sample as the behaviours

Stated

R10 (2024): trust in Ministry of Health; placebo police, courts

R9 (2021-23): trust govt on COVID vaccine safety

Experienced

R10: difficulty, bribes, absent staff, waiting

R9: felt respected by clinic staff

Lived

Fieldwork: why people use, bypass or avoid clinics

Trust in biomedicine, the health system and health workers (Gilson 2003)

Backup: the instrument and its limits

Instrument

Medical propensity of the station's society, measured in other territories (leave-one-out)

Threat

Societies also differ in doctrine and schooling, which can affect health directly

Response

Control for denomination and schools

Balance tests across societies

Plausibly exogenous bounds (Conley, Hansen and Rossi 2012)

The instrument is one strand of evidence, not the whole case.

Backup: mediation, stated honestly

1. Primary result: the total effect of mission medical exposure
2. Controlled direct effect via sequential g-estimation (Acharya, Blackwell and Sen 2016)
3. Sensitivity analysis: how much hidden confounding erases the trust share (Imai, Keele and Tingley 2010)
4. Common geography: DHS links are individual-level; Afrobarometer and burden at district or cluster level
5. Triangulate with the Zambian fieldwork

Backup: risks and mitigations

Risk	Mitigation
Uneven archival coverage	Catalogue audit before travel; richest territories first; results hold on the core set
Selection of mission sites	Layered design (slide 7); sensitivity bounds
Data access (DHS, Afrobarometer)	Download DHS rounds and GPS early; IPUMS copies; district-level fallback
Approvals in Zambia	Applications from month 1; fieldwork starts month 20
Travel or photography limits	British Online Archives; archive copying services

Backup: outcomes and exposure

Primary outcomes (DHS)

Child mortality, vaccination, antenatal care, facility births, care-seeking

Malaria test and anaemia results where collected

Secondary (IHME)

Local Burden of Disease maps, with uncertainty carried through

Malaria: care-dependent outcomes, not prevalence

Exposure

Distance; 10 and 25 km bands; years open; patients per head

Bands wider than DHS displacement (2 km urban, 5 km rural)

Backup: telling the channels apart

Channel	What you'd see	Main evidence
Surviving faith-based facilities	Effect concentrated where the mission facility still runs	KEMRI-WHO faith-based flag; CHAZ list; DHS place of delivery
Health-worker norms	Survives closure; clinic care follows the mission's history	Training records; DHS and Afrobarometer care-quality items
Religious norm	Strongest among members of the mission's church	Religion in DHS and Afrobarometer; church teaching
Coercion	Lower trust in medicine even where care is no worse	Archival coding; health vs police trust (Afrobarometer); memories

Care quality: better after mainly care, possibly harsher after coercion. The fieldwork shows why.

Backup: budget at a glance

Budget line	ZAR
PI salary, 48 months at 100%	4,799,860
Research expenses (RA, fieldwork, travel, open access, equipment, other)	1,270,054
of which RA (24 months) / fieldwork / travel	325,939 / 500,190 / 182,500
Overheads (20% of direct costs)	1,213,983
Total requested	7,283,897